Research Literature & Design Insights

08 September 2026 11:16

Paper 1

Swacha & Gracel (2025), "Retrieval-Augmented Generation (RAG) Chatbots for Education: A Survey of Applications."

Pages 1–3 — Key Understanding

• LLM chatbots can provide accessible conversational learning support.
• Students may benefit from asking questions without social pressure.
• Major LLM problems for education:
  - Problem 1 — Difficult to update or extend knowledge
  - Problem 2 — Hallucination

• So what does RAG do?
The paper says RAG improves an LLM **not by changing its training**, but by adding an **information retrieval step**.

Input x
↓
Retrieve documents z
↓
Use z as additional context
↓
Generate answer y

• RAG can still fail:
  - Retriever may retrieve irrelevant information.
  - LLM may fail to use retrieved information properly.
  - Retrieval + generation adds computational overhead.

Pages 4–8— Key Understanding

Key findings:

• Survey analyzed 47 educational RAG papers.
• Publications included conference papers, journal articles, preprints and Master's theses.
• Most identified works were published in 2024.

Educational RAG chatbot purposes were classified into:
1. Learning
2. Access to source knowledge
3. Access to assessment information
4. Organizational matters
5. Local culture
6. Course selection
7. Question generation
8. Learning analytics support

• Access to source knowledge was the largest category (20 papers).
• Learning was the second-largest category (14 papers).
• Learning-oriented systems add features such as personalization, adaptive support, question/exercise generation, answer evaluation, follow-up questioning, mind maps, reports and traceability.

Thematic scope:
• Institutional-domain knowledge: 11 papers
• Health sciences: 9
• Computer science: 6
• Other areas included law, education, math, finance, language, biomedical, culture and multi-domain/any-domain systems.

Computer Science examples included databases, programming, data science/AI and access to CS literature.

LLMs:
• GPT appeared in 36 instances.
• LLaMA appeared in 15 instances.
• Other LLMs appeared in 8 instances.
• 7 instances did not identify the LLM.
• These are model-use instances, not necessarily unique papers, because papers comparing multiple models were counted multiple times.

Relevance to OSTutorLLM:
• OSTutorLLM belongs primarily to Learning + Access to Source Knowledge.
• Question generation, assessment support and learning analytics
are possible future extensions.
• Our domain is Computer Science → Operating Systems.
• We should not claim that GPT is automatically the best LLM just
because it is the most frequently used.
• Final LLM selection should be based on controlled evaluation.

Pages 9–12
Evaluation of Educational RAG Chatbots

• RAG performance depends on multiple components:
retrieval model, LLM, corpus, prompt and tuning.
Good individual components do not guarantee a good final system.

• There is no widely accepted standard for evaluating RAG chatbots.

Evaluation dimensions include:

1. Retrieval evaluation
- Precision
- Recall
- F1
- Accuracy
- Sensitivity
- Specificity

2. LLM evaluation
- LLM benchmarks such as AGIEval

3. Conversational quality
- Correctness
- Completeness
- Communication
- Relevance
- Clarity
- Credibility

4. Generated text quality
- BERTScore
- BLEURT
- ROUGE
- Fluency
- Coherence
- Readability

5. Software usability
- Response time
- User acceptance
- User satisfaction
- SUS
- Safety

6. Economic evaluation
- Cost per query
- Cost per student

RAGAS:
• Faithfulness — is the answer grounded in retrieved context?
• Answer Relevance — does the answer address the question?
• Context Relevance — is the retrieved context relevant?

Educational evaluation can additionally consider:
• Learning success
• Motivation
• Usability
• Acceptance
• Psychological factors
• Technical correctness
• Other beneficial effects

Important insight:
A RAG system can be technically good without necessarily
improving student learning.

For OSTutorLLM:

We should eventually evaluate at multiple levels:
1. Retrieval quality
2. Answer/RAG quality
3. Educational effectiveness
4. Possibly usability, speed and cost

Human + automatic evaluation can be combined.

Pages 13–16
Discussion, Recommendations & Conclusions

Major findings:
• Educational RAG is still a relatively new research area.
• 47 relevant papers were identified.
• Most systems primarily support students.
• Computer Science was the second most represented area-specific domain after health sciences.
• GPT was the most frequently used LLM family.
• There is no single comprehensive evaluation method for educational RAG chatbots.

MOST IMPORTANT RESEARCH GAP:
• None of the studies analyzed in the survey reported the actual educational effect of the chatbot for its intended goal.
• Technical quality alone does not prove educational effectiveness.

For learning chatbots, the authors recommend:
• Pre-test students before chatbot-supported learning.
• Let an experimental group use the chatbot.
• Use a control group that does not use the chatbot.
• Conduct a subject-specific post-test.
• Compare learning outcomes between groups.

For source-knowledge chatbots:
• Compare project/exam grades between chatbot users and a control group.

Three major RAG factors:
1. Underlying LLM
2. Source content
3. Vector representation/retrieval approach

Other considerations:
• Privacy and plagiarism risks must be considered.
• Ready-made RAG components make development easier but do not remove project risks.
• Most important evaluation = whether the system achieves its intended goal.
• Technical and financial constraints must also be considered.

Paper limitations:
• Only three bibliographic sources.
• Search terms may have missed some relevant studies.
• Publication bias may favor successful systems.
• Only English publications were included.

Relevance to OSTutorLLM:
• We should evaluate both technical RAG performance and actual OS learning effectiveness.
• Potential future experiment:
  Pre-test → OSTutorLLM learning → Post-test compared against a control group.
• Our project should not be presented merely as "a chatbot"; it should be presented as an educational OS tutoring system.